School of Future Tech

Case Study Report

on

Movie Ticket Booking System (C++)

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1. INTRODUCTION TO CASE STUDY:

Movie theatres handle a large number of customers daily who want to book tickets for different movies and shows. Traditionally, ticket booking was done manually at counters, which often caused long queues, booking errors, and inefficient seat management.

With the advancement of computer applications, booking systems can be automated to simplify ticket management. A **Movie Ticket Booking System** helps users view available movies, select show timings, book seats, and generate a bill automatically.

This case study demonstrates the development of a **menu-driven Movie Ticket Booking System using C++ programming language**. The system simulates the ticket booking process in a theatre using basic programming constructs such as **arrays, loops, structures, and conditional statements**.

The application runs on a **console interface** and allows users to interact with the system through a simple menu. It also calculates the total ticket cost based on seat selection and ticket price.

2. PROBLEM STATEMENT / CASE BACKGROUND (ABSTRACT):

The purpose of this case study is to design and implement a Movie Ticket Booking System using C++ that allows users to perform basic ticket booking operations.

The system provides functionality to:

- Display available movies
- Select a movie show
- Book seats
- Calculate the total ticket price
- Generate a bill for the user

The application focuses on implementing fundamental programming concepts in C++ such as:

- Arrays
- Loops
- Conditional statements
- Functions

The system acts as a simplified simulation of a real-world movie theatre booking process.

3.CASE STUDY DESIGN:

The Movie Ticket Booking System is designed as a **menu-driven console application**.

The program is divided into several modules that handle different operations within the system.

Main Modules

1. Movie Display Module

Displays a list of available movies along with their show timings and ticket prices.

Example:

2. Ticket Booking Module

Allows users to:

- Select a movie
- Enter the number of seats
- Confirm booking

3. Billing Module

After booking seats, the system calculates:

Total Cost = Number of Seats × Ticket Price

The system then displays the final bill.

4. Exit Module

Allows users to safely exit the program.

4. METHOD AND ALGORITHMS APPLIED TO CASE STUDY:

The Movie Ticket Booking System applies several basic programming techniques in C++.

1. Arrays

Arrays are used to store movie information such as:

- Movie names
- Show timings
- Ticket prices

2. Conditional Statements

Conditional statements (`if`, `switch`) are used to control program flow.

Example uses:

- Checking user menu choice
- Validating movie selection
- Calculating total price

3. Loops

Loops are used to:

- Display the menu repeatedly
- Display movie lists
- Allow multiple bookings

5. CASE STUDY IMPLEMENTATION DETAILS AND SNAPSHOTS:

The program is implemented using C++ and runs in a console environment.

Step 1: Display Movie List

When the user selects View Movies, the system displays all available movies.

Step 2: Ticket Booking

When the user selects **Book Ticket**, the system asks for:

- Movie selection
- Number of seats

Step 3: Bill Calculation

The system calculates the total ticket cost.

Step 4: Display Bill

Step 5: Exit Option

Users can exit the program by selecting option 3.

SNAPSHOTS: (CODE)

final case study > ninad.cpp > main()

```cpp
#include <iostream>

using namespace std;

const int ROWS = 5;
const int COLS = 20;
const double GST = 0.18;
const string ADMIN_PASS = "admin123";

struct Seat {
    bool booked = false;
};

struct Show {
    string movieName;
    string timing;
    vector<vector<Seat>> seats;
    double revenue = 0;
};

vector<Show> shows;

string getClass(char row) {
    if (row == 'A') return "Platinum";      (const char [5])"Gold"
    if (row == 'B' || row == 'C') return "Gold";
    return "Silver";
}

double getBasePrice(string seatClass, bool weekend) {
    if (seatClass == "Platinum") return weekend ? 500 : 400;
    if (seatClass == "Gold") return weekend ? 350 : 300;
    return weekend ? 250 : 200;
}

void initializeShows() {
    Show s1 = {"Avengers", "2:00 PM", vector<vector<Seat>>(ROWS, vector<Seat>(COLS))};
    Show s2 = {"Oppenheimer", "6:00 PM", vector<vector<Seat>>(ROWS, vector<Seat>(COLS))};
    Show s3 = {"Dune 2", "9:00 PM", vector<vector<Seat>>(ROWS, vector<Seat>(COLS))};
    shows.push_back(s1);
    shows.push_back(s2);
    shows.push_back(s3);
}

void displaySeats(Show &show) {
    cout << "\nSeat Layout (X = Booked)\n\n";
```

final case study > ninad.cpp > main()

```cpp
void bookTickets() {
}

void adminPanel() {
    string pass;
    cout << "Enter Admin Password: ";
    cin >> pass;

    if (pass != ADMIN_PASS) {
        cout << "Wrong Password!\n";
        return;
    }

    cout << "\n--- Admin Dashboard ---\n";
    double totalRevenue = 0;
    for (auto &s : shows) {
        cout << s.movieName << " Revenue: ₹" << s.revenue << endl;
        totalRevenue += s.revenue;
    }
    cout << "Total Revenue: ₹" << totalRevenue << endl;
}

int main() {
    srand(time(0));
    initializeShows();

    int choice;
    do {
        cout << "\n🎬 SMART MULTIPLEX MANAGEMENT SYSTEM 🎬\n";
        cout << "1. Book Tickets\n";
        cout << "2. Admin Panel\n";
        cout << "3. Exit\nChoice: ";
        cin >> choice;

        switch(choice) {
            case 1: bookTickets(); break;
            case 2: adminPanel(); break;
            case 3: cout << "Thank You!\n"; break;
            default: cout << "Invalid Option!\n";
        }
    } while (choice != 3);

    return 0;
}
```

OUTPUT

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
cd "/Users/ninadnileshdeodhare/Desktop/Project/C++ Main/" && g++ movie.cpp -o movie && "/Users/ninadnileshdeodhare/Desktop/Project/C++ Main/"movie
ninadnileshdeodhare@Ninads-MacBook-Air Project % cd "/Users/ninadnileshdeodhare/Desktop/Project/C++ Main/" && g++ movie.cpp -o movie && "/Users/ninadnilesh
deodhare/Desktop/Project/C++ Main/"movie

🎬 SMART MULTIPLEX MANAGEMENT SYSTEM 🎬
1. Book Tickets
2. Admin Panel
3. Exit
Choice:
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
cd "/Users/ninadnileshdeodhare/Desktop/Project/C++ Main/" && g++ movie.cpp -o movie && "/Users/ninadnileshdeodhare/Desktop/Project/C++ Main/"movie
ninadnileshdeodhare@Ninads-MacBook-Air Project % cd "/Users/ninadnileshdeodhare/Desktop/Project/C++ Main/" && g++ movie.cpp -o movie && "/Users/ninadnilesh
deodhare/Desktop/Project/C++ Main/"movie

🎬 SMART MULTIPLEX MANAGEMENT SYSTEM 🎬
1. Book Tickets
2. Admin Panel
3. Exit
Choice: 1

Available Shows:
1. Avengers - 2:00 PM
2. Oppenheimer - 6:00 PM
3. Dune 2 - 9:00 PM
Select Show: 2

Seat Layout (X = Booked)

A 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
B 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
C 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
D 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
E 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Number of tickets: 2
Is it weekend show? (1-Yes 0-No): 0
Enter Seat (A 5): A 1
Enter Seat (A 5): A2

Food Menu:
1.Popcorn(200)
2.Combo(350)
3.None
Choice: 2
Quantity: 1

Payment Mode:
1.UPI
2.Card
3.Cash
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                   Code - C++ Main

ninadnileshdeodhare@Ninads-MacBook-Air Project % cd "/Users/ninadnileshdeodhare/Desktop/Project/C++ Main/" && g++ movie.cpp -o movie && "/Users/ninadnilesh
deodhare/Desktop/Project/C++ Main/"movie
1.UPI
2.Card
3.Cash
Choice: 3
Processing Payment...
Payment Successful!

===== BOOKING CONFIRMED =====
Booking ID: BK85852
Movie: Oppenheimer
Show: 6:00 PM
Seats: A1 A2
Subtotal: ₹1150
GST: ₹207
Total Paid: ₹1357
============================

🎬 SMART MULTIPLEX MANAGEMENT SYSTEM 🎬
1. Book Tickets
2. Admin Panel
3. Exit
Choice: 2
Enter Admin Password: admin123

--- Admin Dashboard ---
Avengers Revenue: ₹0
Oppenheimer Revenue: ₹1357
Dune 2 Revenue: ₹0
Total Revenue: ₹1357

🎬 SMART MULTIPLEX MANAGEMENT SYSTEM 🎬
1. Book Tickets
2. Admin Panel
3. Exit
Choice: 3
Thank You!
ninadnileshdeodhare@Ninads-MacBook-Air C++ Main %
```

6. CASE STUDY RESULTS AND CONCLUSION:

The Movie Ticket Booking System successfully demonstrates the implementation of a basic theatre booking process using C++.

The system allows users to:

- View available movies
- Select a movie show
- Book tickets
- Calculate total bill automatically

Future Improvements:

The system can be enhanced by:

- Adding seat layout visualization
- Implementing seat availability tracking
- Adding food ordering system
- Integrating database storage
- Creating a GUI or web-based application

7. REFERENCES:

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3. GeeksforGeeks – https://www.geeksforgeeks.org
4. Introduction to Programming with C++